COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_piecewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit}; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, ∼0.3 m	measured / conservative
$l_{p,\phi}$, $l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	conservative geometry
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T\Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
$\varepsilon_{\max}$	ramp-slope gate	3%	sensitivity chain (Sec. 4)
$N_{\min}$	window-sample gate	38	30 pre + 8 post
$e_{\text{lin,max}}$	linearity-RMSE gate	30 mN m	10× noise floor
N_{seg}	sweep margin (each end)	8	degenerate-candidate guard

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 6).

2 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos\varphi \approx 1$, $\sin\varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P\,\dot\omega = M(t) + f\,l_p - W\,(l_p + \lambda_{\text{off}}) + W z_{\text{CoM}}\,\varphi, \tag{1}$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\text{crit}}) = 0, \qquad \dot\omega(t_{\text{crit}}) = 0. \tag{2}$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\text{crit}} + \dot M\,\tau$ in onset-relative time $\tau = t - t_{\text{crit}}$, the exact solution of (1)–(2) collapses to the hyperbolic closed form

$$\boxed{\omega(\tau) = C_1\big(\cosh(C_2\tau) - 1\big), \qquad C_1 = K\dot M, \quad K = \frac{1}{W z_{\text{CoM}}}, \quad C_2 = \sqrt{\frac{W z_{\text{CoM}}}{J_P}}} \tag{3}$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = W z_{\text{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as rig constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \to 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

3 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i}\, \Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\text{end}} = \arg\max_t |M|$, *truncated at the first sample exceeding the allocator cap* M_{cap}: past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

4 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\text{cmd}})/\dot{M}_{\text{cmd}}$ and $\dot{M}$ the least-squares slope over the window. A 3% error in $\dot{M}$ perturbs $C_1 = K\dot{M}$ by 3% — about 7× below the ±20% level the sensitivity analysis shows harmless.
2. **Window samples** $N_{\text{full}} \geq 38$: a necessary condition for any onset position. Because the window opens at $|M| = 10$ mN m while $M_{\text{crit}} \gtrsim 0.4$ N m, the onset cannot occur earlier than $\hat{N}_{\text{pre}} = M_{\text{crit}}^{\min}/(\dot{M}\, T_s) \geq 30$ samples into the window even at the fastest ramp; adding the $N_{\text{seg}} = 8$ post-onset fit minimum gives 38.
3. **Linearity RMSE** ≤ 30 mN m about the run's own best-fit line: a fault detector for aborted/stepped ramps, set at 10× the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K\dot{M}$.

5 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Algorithm 1 COSH onset detection (single run)

Require: samples $\{(t_i, \omega_i)\}$ on the excitation window $[i_{\text{start}}, i_{\text{end}}]$; moment $M(t)$; calibrated (C_2, K); measured ramp slope $\dot{M}$; margin $N_{\text{seg}} = 8$

Ensure: onset index j^*; critical moment M_{crit}

```
C_1 ← K Ṁ                                              ▷ onset conditions (2) fix the amplitude
for j = i_start + N_seg, ..., i_end − N_seg do          ▷ exhaustive sweep of the single unknown
    C ← median{ω_i : i_start ≤ i < j}                   ▷ baseline by continuity
    cost(j) ← Σ_{i=i_start}^{j−1} (ω_i − C)^2 + Σ_{i=j}^{i_end} (ω_i − [C_1(cosh C_2(t_i − t_j) − 1) + C])^2
end for
j* ← arg min_j cost(j);    M_crit ← M(t_{j*})
```

Two implementation details. *Baseline by median.* The pre-onset gyro trace is heavy-tailed (propeller vibration, spin-up spikes), and — more importantly — when a candidate j overshoots the true onset, the pre-segment tail contains the beginning of the hyperbolic rise. A mean baseline drifts with that leaked rise and partially cancels the residual it should expose, flattening the cost landscape; the median stays at the rest level until half the segment is contaminated, so misplaced candidates are penalized sharply and the minimum stays well defined. It is also a one-shot, seedless statistic, consistent with the no-optimizer design. An ablation rerunning the full pipeline — including the (C_2, K) calibration — with the mean instead (`analysis/baseline_stat_ablation.py`) leaves the run gating unchanged and 62/132 onsets identical, but biases the remainder toward *earlier* onsets, shrinking the directional means in magnitude by up to 30 mN m on the noisiest datasets. The route is indirect. At *fixed* (C_2, K) a baseline biased toward the motion direction *delays* the detected onset: the post-onset residual $C_1[\cosh(C_2(t - t_{\text{crit}}^{\text{real}})) - \cosh(C_2(t - t_{\text{crit}}^{\text{est}}))] - C$ can only vanish with a later split, and an oppositely biased baseline advances it. The dominant effect, however, is that the leaked-rise flattening of the cost landscape moves the CV-selected calibration to a different point on the (C_2, K) ridge — e.g. $C_2 : 8.00 \to 5.38$, $K : 0.060 \to 0.150$ on the most affected dataset — whose net effect is the earlier-onset magnitude shrink; the ramp-invariance CV itself is essentially unchanged. The pivot-free offset M_{ff} moves by at most 4.2 mN m (0.13 mm), well below the run-to-run scatter — the median removes a systematic bias and tunes nothing. A second ablation pins $C = 0$, the ideal-sensor value implied by the onset conditions: the measured pre-onset baseline is in fact nonzero (gyro bias plus residual spin-up motion; $|C|$ median 3.1, p90 14.2, max 30.6 mrad/s), yet the estimate barely moves (71/132 onsets identical, M_{ff} shift ≤ 1.3 mN m). The estimated baseline is therefore not load-bearing on this dataset; it is kept because it makes the sweep invariant to the gyro bias — a rig- and temperature-dependent quantity that need not stay this small — at zero parameter cost. Nor does C need the bias to be constant in time: it is re-estimated per run, so only *within-window* drift (windows last 1–10 s) matters. Measured over the guaranteed pre-onset segment ($|M| < 0.3$ N m, spans ≤ 3.0 s), the baseline wanders by 3.4 mrad/s (median; p90 24, max 68) — an upper bound that conflates true sensor drift with real elastic pre-onset motion (`analysis/bias_drift.py`). This wander is present in the very data on which the $C = 0$ bound and the rate-invariance certificates were earned, and its structure is covered by the perturbation absorption: the constant part goes to C, a τ-linear trend is absorbed by the span projection, and only the residual curvature reaches ρ. *Sweep margin.* The margin $N_{\text{seg}} = 8$, applied at both window ends, guarantees every candidate split leaves at least N_{seg} samples in each segment; it excises only degenerate candidates: near $j = i_{\text{start}}$ the baseline would be estimated from a handful of samples, and near $j = i_{\text{end}}$ the post-onset arc is too short for the cosh branch to be falsifiable (at $j = i_{\text{end}}$ its cost is nearly an empty sum). Since physically feasible onsets lie at least $\hat{N}_{\text{pre}} \geq 30$ samples inside the window (Sec. 4) and the window closes only after the response is well developed, the sweep remains exhaustive over every feasible onset.

Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a* prediction test*, not flexibility: across all gate-passing runs the normalized post-onset residual is* $\approx 6\%$ *(gyro-noise dominated) and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

6 Rig-constant calibration

Algorithm 1 presupposes (C_2, K). Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the ground-effect spring). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion, and the reason can be made exact. The identity $\cosh x - 1 = 2\sinh^2(x/2)$ factors the closed form as

$$C_1\big(\cosh(C_2\tau) - 1\big) \;=\; \underbrace{\tfrac{C_1}{2}\, e^{-C_2 t_{\text{on}}}}_{A}\; e^{C_2 t}\,\big(1 - e^{-C_2\tau}\big)^2, \tag{4}$$

whose last factor rises to one within a few growth times C_2^{-1}. On the tail the amplitude and the onset therefore enter the data only through the product A, and the substitution

$$C_1 \to \lambda C_1, \qquad t_{\text{on}} \to t_{\text{on}} + C_2^{-1}\ln\lambda \tag{5}$$

leaves A *exactly* invariant for every $\lambda > 0$: the response changes only through the last factor, by the relative amount $\big[(1 - \lambda e^{-C_2\tau})/(1 - e^{-C_2\tau})\big]^2 - 1 \approx -2(\lambda - 1)\, e^{-C_2\tau}$, which decays exponentially in τ. An earlier onset with a smaller amplitude thus trades off against a later onset with a larger one everywhere except near the onset — where the response is quadratic, $\frac{1}{2}C_1 C_2^2\tau^2$, and of the order of the gyro noise. Equivalently, the sensitivity vectors $\partial\omega/\partial C_1 \propto \cosh(C_2\tau) - 1$ and $\partial\omega/\partial t_{\mathrm{on}} \propto \sinh(C_2\tau)$ become parallel on the tail, so the Gauss–Newton Hessian of the summed residual is nearly singular along $\delta C_1/C_1 = C_2\,\delta t$ — a *shallow ridge* in the (C_2, K) plane whose minimizer is set by noise, not physics. This is an ill-conditioning statement, not a non-identifiability one: in the noiseless limit the post-onset shape determines all three parameters uniquely. Under gyro noise, however, the information along (5) is exponentially small, so per-run estimates — and with them the detected critical moments — scatter along the ridge. The calibration should therefore be read as *variance suppression*: choosing (C_2, K) once per dataset to minimize the dispersion of M_{crit} across ramp rates removes precisely that scatter, which is where the practical strength of the method lies.

The ridge is invisible to the residual but not to the deliverable. Moving along (5) shifts every detected onset by the same *time* offset $\delta t = C_2^{-1}\ln\lambda$ — with $\lambda = K^*/K$ the same for all runs, since $C_1 = K\dot{M}$ scales them alike — and therefore biases the critical moment by

$$\delta M_{\mathrm{crit}} \;=\; \dot{M}\,\delta t \;\propto\; \dot{M}, \tag{6}$$

linearly in the ramp rate: a wrong pair *fans out* the identified M_{crit} across the twelvefold rate range, while the correct pair leaves it rate-invariant. The direction the residual cannot see is thus exactly the direction a rate-consistency criterion sees best. We therefore calibrate by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \ldots, 1.20\}$ N m/s,

$$\mathrm{score}(C_2, K) \;=\; \sum_{\mathrm{dir}\in\{+,-\}} \mathrm{CV}\Big(\big\{M_{\mathrm{crit}}^{(r)}(C_2, K)\big\}_r\Big), \qquad \mathrm{CV} = \mathrm{std}/|\mathrm{mean}|, \tag{7}$$

where $M_{\mathrm{crit}}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r.

Search ranges. The admissible box is set so that it covers every physically plausible constant with margin. For $K = 1/(W z_{\mathrm{CoM}})$, the coarse range $K \in [0.10, 0.50]$ $(\mathrm{N\,m})^{-1}$ corresponds to $z_{\mathrm{CoM}} \in [0.06, 0.32]$ m at the vehicle weight — bracketing the plausible CoM heights — with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ (z_{CoM} up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(KC_2^2)$ spans roughly $[0.03, 1.1]$ $\mathrm{kg\,m^2}$, bracketing the parallel-axis estimate $J + m z_{\mathrm{CoM}}^2 \approx 0.2$–$0.35$ $\mathrm{kg\,m^2}$ by a factor of several on both sides.

Algorithm 2 Ramp-invariance consistency score

Require: gated runs at rates $r \in \{0.10, \ldots, 1.20\}$ N m/s, both tip directions; candidate pair (C_2, K)
Ensure: consistency score s ▷ lower = M_{crit} more rate-invariant
for each gated run r **do**
  $M_{\mathrm{crit}}^{(r)} \leftarrow$ Algorithm 1 with (C_2, K) pinned ▷ stride-3 sweep
end for
for dir $\in \{+,-\}$ **do**
  $\mathrm{CV}_{\mathrm{dir}} \leftarrow \mathrm{std}\big(\{M_{\mathrm{crit}}^{(r)}\}_{r\in\mathrm{dir}}\big) \,/\, \big|\mathrm{mean}\big(\{M_{\mathrm{crit}}^{(r)}\}_{r\in\mathrm{dir}}\big)\big|$
end for
return $s \leftarrow \mathrm{CV}_+ + \mathrm{CV}_-$ ▷ Eq. (7): a static threshold must not depend on the approach rate

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Algorithm 3 Rig-constant calibration (coarse-to-fine)

Require: gated runs; box $C_2 \in [3, 8]$ rad/s, $K \in [0.05, 0.70]$ $(\mathrm{N\,m})^{-1}$
Ensure: calibrated pair (C_2^*, K^*)
coarse: evaluate Algorithm 2 on the $11{\times}11$ grid $C_2 = 3.0{:}0.5{:}8.0$, $K = 0.10{:}0.04{:}0.50$
$(C_2^b, K^b) \leftarrow \arg\min$ over the coarse grid
fine: evaluate Algorithm 2 on a $9{\times}9$ uniform grid over $[C_2^b \pm 0.5] \times [K^b \pm 0.04]$, clipped to the box ▷ resolution 0.125 rad/s, 0.01 $(\mathrm{N\,m})^{-1}$
$(C_2^*, K^*) \leftarrow \arg\min$ over the fine grid

Coarse-to-fine procedure. Algorithm 2 states the consistency criterion as a stand-alone routine — it wraps Algorithm 1 and returns the directional sum of coefficients of variation — and Algorithm 3 performs the search over it. The coarse scan evaluates (7) on an 11×11 grid ($\Delta C_2 = 0.5$ rad/s, $\Delta K = 0.04$ $(\mathrm{N\,m})^{-1}$;

121 evaluations). A single refinement pass then re-scans a 9×9 uniform grid spanning $\pm$ one coarse step about the coarse minimizer (± 0.5 rad/s, ± 0.04 $(\mathrm{N\,m})^{-1}$, clipped to the box), i.e. final resolutions of 0.125 rad/s and 0.01 $(\mathrm{N\,m})^{-1}$ — so the reported constants are not quantized to the coarse step. One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found to cost about $4\times$ more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

Remark 2 *The objective (7) retains a shallow valley, so the individual constants are only loosely determined: $W z_{\mathrm{CoM}} = 1/K$ and $J_P = 1/(KC_2^2)$ are order-of-magnitude sanity checks, not precision measurements. What the criterion pins down — by construction — is the identified onset: M_{crit} is robust along the valley, and a $\pm 20\%$ mis-specification of either constant moves the identified CoM by less than* 0.9 *mm.*

7 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\mathrm{ff}} = \tfrac{1}{2}\,(\bar{M}_+ + \bar{M}_-), \tag{8}$$

in which the pivot-arm terms $\pm(W - f)\,l_p$ — and any thrust-proportional contamination acting through the arm l_p, ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight. A roll excitation (M_x) senses the y-offset via $M_{\mathrm{ff},x} = +W y_{\mathrm{off}}$; a pitch excitation ($M_y$) senses the x-offset via $M_{\mathrm{ff},y} = -W x_{\mathrm{off}}$.

8 Comparison with per-run estimators: NLS, PELT, CUSUM

Four per-run alternatives were run on *identical* inputs — same signals, excitation windows, allocator caps and onset-free gates (the gated run set is method-independent because the gates use the moment trace only), 132 runs in total (`analysis/nls_comparison.py`):

- **NLS** — the predecessor of Algorithm 1: per-run nonlinear least squares (SCIPY `least_squares`, trust-region reflective), free (C_1, C_2, C) with C_2 bounded to $[3, 8]$ rad/s, onset swept locally around a quadratic-model seed;
- **PELT$_{\mathrm{N}}$ / PELT$_{\mathrm{R}}$** — change-point detection (binary segmentation, `ruptures`) with Gaussian mean+variance and RBF-kernel costs;
- **CUSUM** — the tabular sequential detector with allowance $k = 2\sigma$ against the pre-onset vibration.

Figure 1 is the readable summary: per case–axis pair it shows each estimator's CoM offset with its 95% confidence interval (Welch t on the pivot-free average, run-to-run scatter) against the load-cell truth. The full data sit in the tables: Table 2 compares the deliverable — the CoM offset from the pivot-free average (8) — against the load-cell ground truth; Table 3 lists the underlying directional critical-moment means, pivot-free offsets and their confidence intervals; and Table 4 compares the ramp-rate dispersion of the directional critical moments.

Three observations. *(i) The deliverable is detector-agnostic:* against the load-cell truth all five estimators land at RMS 1.6–1.8 mm (Table 2, bottom rows) — the pivot-free average (8) cancels each method's largely common-mode onset bias between the + and − directions, and 5–7 runs per direction average the rest. That the final CoM offset does not depend on the detection methodology is itself a validation certificate — and Fig. 2 sharpens it: the residual errors that do remain (cases 3–4) are shared by all five estimators, placing them in the rig/contact geometry rather than in any onset detector. *(ii) Run-level robustness is where the methods separate:* COSH has the lowest CV in 16/20 direction groups (PELT$_{\mathrm{N}}$ takes the other 4), median 2.4% against 3.9–7.1%, and its worst group is 6.5% where the alternatives

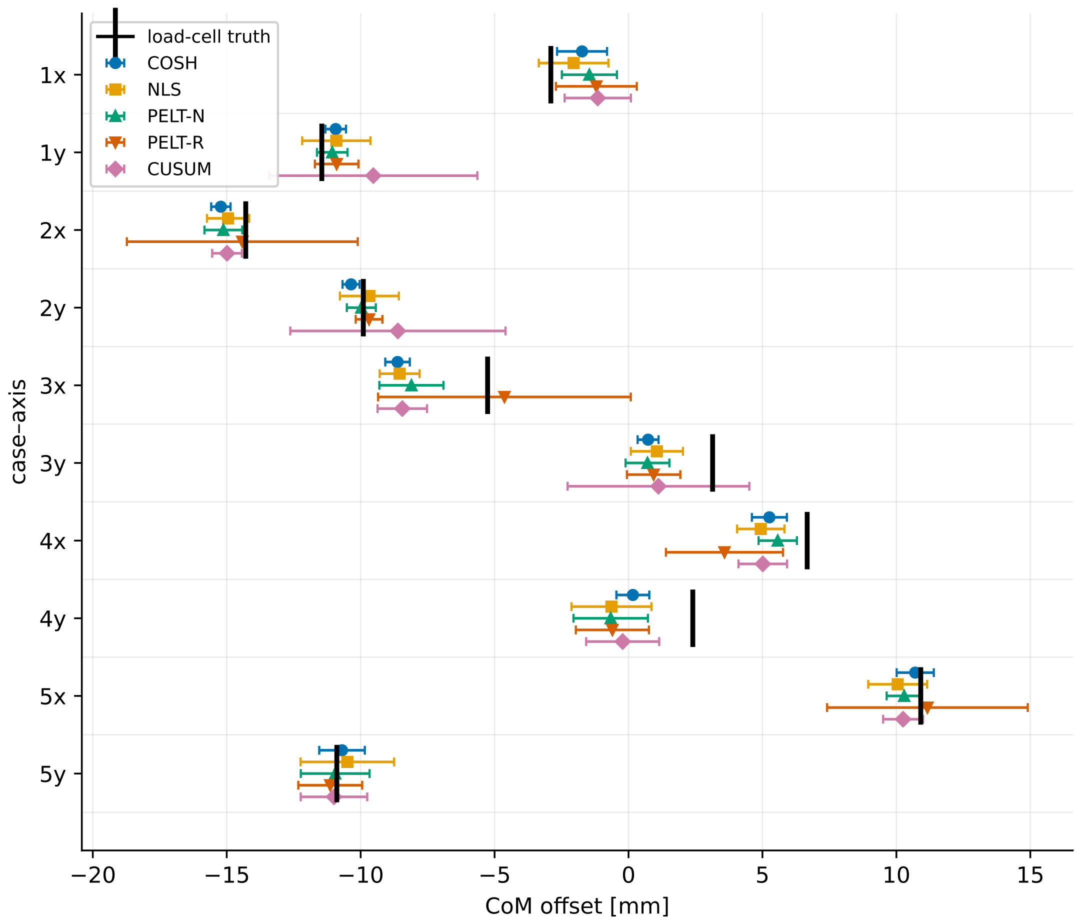

Figure 1: Estimated CoM offset with 95% confidence intervals for the five estimators, against the load-cell truth (black tick), per case–axis pair. COSH (blue) has the narrowest interval in 9 of 10 pairs (±0.3–0.9 mm); the nonparametric detectors blow up to ±4–5 mm on the pairs where they misplace onsets ($\mathrm{PELT_R}$ at 2x, 3x, 5x; CUSUM at 1y, 2y, 3y). Marker shapes duplicate the color coding.

reach 15–27% — the nonparametric detectors ($\mathrm{PELT_R}$, CUSUM) occasionally misplace the onset badly on short fast-ramp windows. Equivalently, the 95% confidence interval on the CoM offset is ±0.3–0.9 mm under COSH — the narrowest in 9 of 10 case–axis pairs — against up to ±4–5 mm for the nonparametric detectors (Fig. 1). The paired per-run critical moments (NLS vs. COSH) differ by 27.6 mN m at the median but up to 217 mN m — the ridge of Sec. 6 realized run by run. *(iii) The biases carry signatures:* NLS sits 2.6% *low* in mean $|M|$ (the early-onset drift of the free amplitude–onset fit), while $\mathrm{PELT_N}$ and CUSUM sit 3.2% and 3.1% *high* — detection lag, since a distribution change must accumulate post-onset evidence before it is declared; both signatures largely cancel in (8). COSH needs no per-run tuning (CUSUM's allowance k, PELT's minimum segment) and a *single* run already sits within a few percent of its directional mean — which is what enables run-count reduction and per-run gating diagnostics.

9 Reproducibility

Every number in this document is reproducible from the repository:

- `critical_value_getter_piecewise.py` — pipeline, gates, caps, calibration, Algorithm 1;
- `analysis/ramp_quality.py` — gates;
- `analysis/sensitivity.py` — ±20% robustness;
- `analysis/cosh_fidelity.py` — prediction-test residuals;

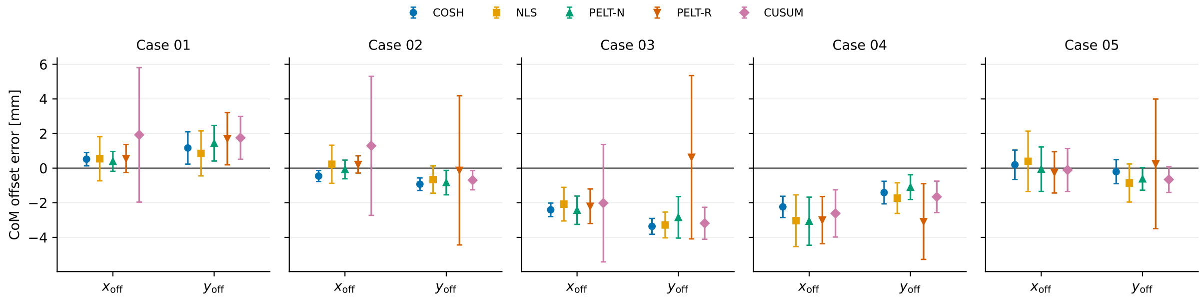

Figure 2: The same data folded about the truth: CoM offset *error* (estimate − truth) with 95% confidence intervals, grouped by component within each case (x_{off} from the pitch excitation, y_{off} from the roll excitation). The residual biases of cases 3 and 4 are *common to all five estimators* — a rig/contact asymmetry, not a detection artifact — while the estimator-to-estimator differences stay within about ±1 mm.

case	axis	λ_{truth} [mm]	estimated CoM offset $\hat{\lambda}\pm$ CI$_{95}$ [mm] COSH	NLS	PELT$_{\text{N}}$	PELT$_{\text{R}}$	CUSUM
1	x	−2.90	−1.73 ±0.93	−2.05 ±1.30	−1.46 ±1.03	−1.20 ±1.51	−1.15 ±1.24
1	y	−11.45	−10.93 ±0.39	−10.91 ±1.27	−11.06 ±0.57	−10.90 ±0.81	−9.53 ±3.88
2	x	−14.29	−15.22 ±0.36	−14.95 ±0.79	−15.13 ±0.71	−14.42 ±4.31	−14.99 ±0.55
2	y	−9.90	−10.36 ±0.32	−9.67 ±1.10	−9.98 ±0.54	−9.69 ±0.50	−8.61 ±4.02
3	x	−5.26	−8.62 ±0.46	−8.54 ±0.74	−8.11 ±1.20	−4.63 ±4.72	−8.45 ±0.92
3	y	+3.14	+0.73 ±0.39	+1.06 ±0.97	+0.71 ±0.82	+0.94 ±1.00	+1.12 ±3.39
4	x	+6.67	+5.26 ±0.65	+4.94 ±0.88	+5.57 ±0.72	+3.58 ±2.19	+5.01 ±0.91
4	y	+2.40	+0.16 ±0.61	−0.64 ±1.49	−0.66 ±1.39	−0.60 ±1.36	−0.22 ±1.36
5	x	+10.91	+10.70 ±0.69	+10.05 ±1.10	+10.29 ±0.65	+11.16 ±3.75	+10.25 ±0.74
5	y	−10.89	−10.70 ±0.85	−10.50 ±1.75	−10.95 ±1.28	−11.14 ±1.19	−11.00 ±1.24
median |err| [mm]			1.05	0.86	0.97	0.59	1.70
RMS |err| [mm]			1.64	1.72	1.67	1.65	1.82
max |err| [mm]			3.36	3.28	3.06	3.09	3.19

Table 2: CoM offset from the pivot-free average (8) against the load-cell ground truth (manuscript Table 7; per-case weight 3.066 kg for case 1, 3.220 kg otherwise), for the five onset estimators on identical inputs. $M_{\text{ff}} = \pm W\hat{\lambda}$ per the axis mapping of Sec. 7.

- `analysis/excitation_angle_design.py` — rate/cutoff design;
- `analysis/ge_linearity.py`, `analysis/ge_trajectory.py` — ground-effect bracket;
- `analysis/tiltcap_ablation.py` — window ablation;
- `analysis/baseline_stat_ablation.py` — median/mean/zero baseline;
- `analysis/bias_drift.py` — in-window baseline drift;
- `analysis/nls_comparison.py` — Tables 2–4.

method		1x	1y	2x	2y	3x	3y	4x	4y	5x	5y
COSH	$\bar{M}_-$	−1.054	−0.422	−1.829	−0.666	−1.542	−0.983	−1.136	−0.998	−1.019	−0.635
	$\bar{M}_+$	+0.950	+1.079	+0.867	+1.320	+0.997	+0.937	+1.468	+0.988	+1.695	+1.310
	M_{ff}	−0.052	+0.329	−0.481	+0.327	−0.272	−0.023	+0.166	−0.005	+0.338	+0.338
	CI_{95}	±0.028	±0.012	±0.011	±0.010	±0.014	±0.012	±0.021	±0.019	±0.022	±0.027
NLS	$\bar{M}_-$	−1.018	−0.399	−1.792	−0.637	−1.573	−0.990	−1.151	−0.928	−0.989	−0.577
	$\bar{M}_+$	+0.894	+1.056	+0.848	+1.248	+1.033	+0.923	+1.463	+0.969	+1.624	+1.241
	M_{ff}	−0.062	+0.328	−0.472	+0.306	−0.270	−0.033	+0.156	+0.020	+0.317	+0.332
	CI_{95}	±0.039	±0.038	±0.025	±0.035	±0.024	±0.031	±0.028	±0.047	±0.035	±0.055
$\mathrm{PELT_N}$	$\bar{M}_-$	−1.048	−0.464	−1.852	−0.703	−1.573	−1.059	−1.164	−1.005	−1.033	−0.640
	$\bar{M}_+$	+0.960	+1.130	+0.896	+1.333	+1.061	+1.014	+1.516	+1.047	+1.683	+1.332
	M_{ff}	−0.044	+0.333	−0.478	+0.315	−0.256	−0.022	+0.176	+0.021	+0.325	+0.346
	CI_{95}	±0.031	±0.017	±0.022	±0.017	±0.038	±0.026	±0.023	±0.044	±0.021	±0.041
$\mathrm{PELT_R}$	$\bar{M}_-$	−1.023	−0.449	−1.680	−0.702	−1.339	−1.055	−1.179	−0.986	−0.932	−0.625
	$\bar{M}_+$	+0.950	+1.105	+0.769	+1.314	+1.046	+0.995	+1.406	+1.024	+1.637	+1.328
	M_{ff}	−0.036	+0.328	−0.455	+0.306	−0.146	−0.030	+0.113	+0.019	+0.353	+0.352
	CI_{95}	±0.045	±0.024	±0.136	±0.016	±0.149	±0.032	±0.069	±0.043	±0.118	±0.038
CUSUM	$\bar{M}_-$	−1.046	−0.480	−1.861	−0.710	−1.608	−1.029	−1.181	−1.044	−1.048	−0.656
	$\bar{M}_+$	+0.977	+1.054	+0.914	+1.254	+1.075	+0.958	+1.498	+1.058	+1.695	+1.351
	M_{ff}	−0.035	+0.287	−0.473	+0.272	−0.267	−0.035	+0.158	+0.007	+0.324	+0.347
	CI_{95}	±0.037	±0.117	±0.018	±0.127	±0.029	±0.107	±0.029	±0.043	±0.024	±0.039

Table 3: Directional critical-moment means $\bar{M}_\pm$, pivot-free offset M_{ff} and its 95% confidence interval (Welch t) [N m] per estimator. Columns are case–axis pairs (e.g. 1x = case 1, roll). Note the signed signatures relative to COSH: NLS magnitudes sit low (early onsets), $\mathrm{PELT_N}$/CUSUM high (detection lag), while M_{ff} stays nearly method-invariant.

		COSH		NLS		$\mathrm{PELT_N}$		$\mathrm{PELT_R}$		CUSUM	
case	axis	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+	CV_-	CV_+
1	x	**3.8**	**5.7**	6.4	7.7	4.4	6.1	7.3	8.6	5.9	6.7
1	y	2.4	**2.2**	8.8	7.2	**1.8**	3.3	5.2	4.4	5.2	24.0
2	x	**1.2**	**1.5**	2.5	4.2	2.5	2.4	15.5	23.6	1.8	2.6
2	y	**1.2**	**1.6**	4.9	5.8	2.8	2.5	3.1	2.0	3.1	21.9
3	x	**1.7**	**2.4**	2.4	4.1	4.9	4.2	23.9	8.5	3.1	4.7
3	y	**1.7**	**2.6**	3.2	7.0	4.4	4.1	4.4	6.0	12.4	22.8
4	x	**3.4**	2.2	3.8	3.5	4.0	**1.6**	5.6	10.3	3.9	3.5
4	y	**2.9**	**3.7**	10.3	6.1	7.5	7.2	7.8	6.9	6.7	7.3
5	x	3.9	2.1	7.2	2.6	**3.6**	**2.0**	27.4	3.0	4.0	2.3
5	y	**6.5**	**2.9**	15.2	5.4	10.1	3.8	9.5	3.8	9.3	4.0
median / worst [%]		**2.4** / **6.5**		5.6 / 15.2		3.9 / 10.1		7.1 / 27.4		4.9 / 24.0	

Table 4: Ramp-rate dispersion (CV [%]) of the directional critical moments under the five estimators. Bold marks the lowest CV in each of the 20 direction groups: COSH in 16, $\mathrm{PELT_N}$ in 4, the rest in none. The underlying directional means are in Table 3.